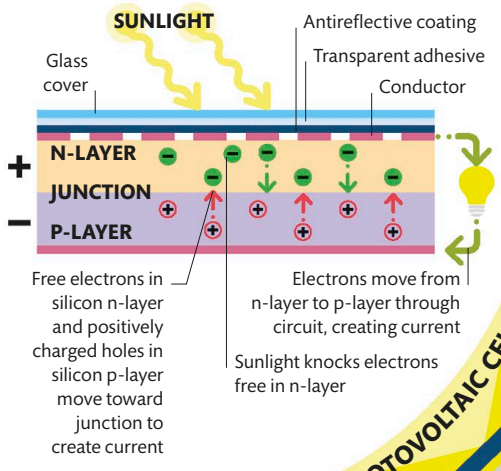
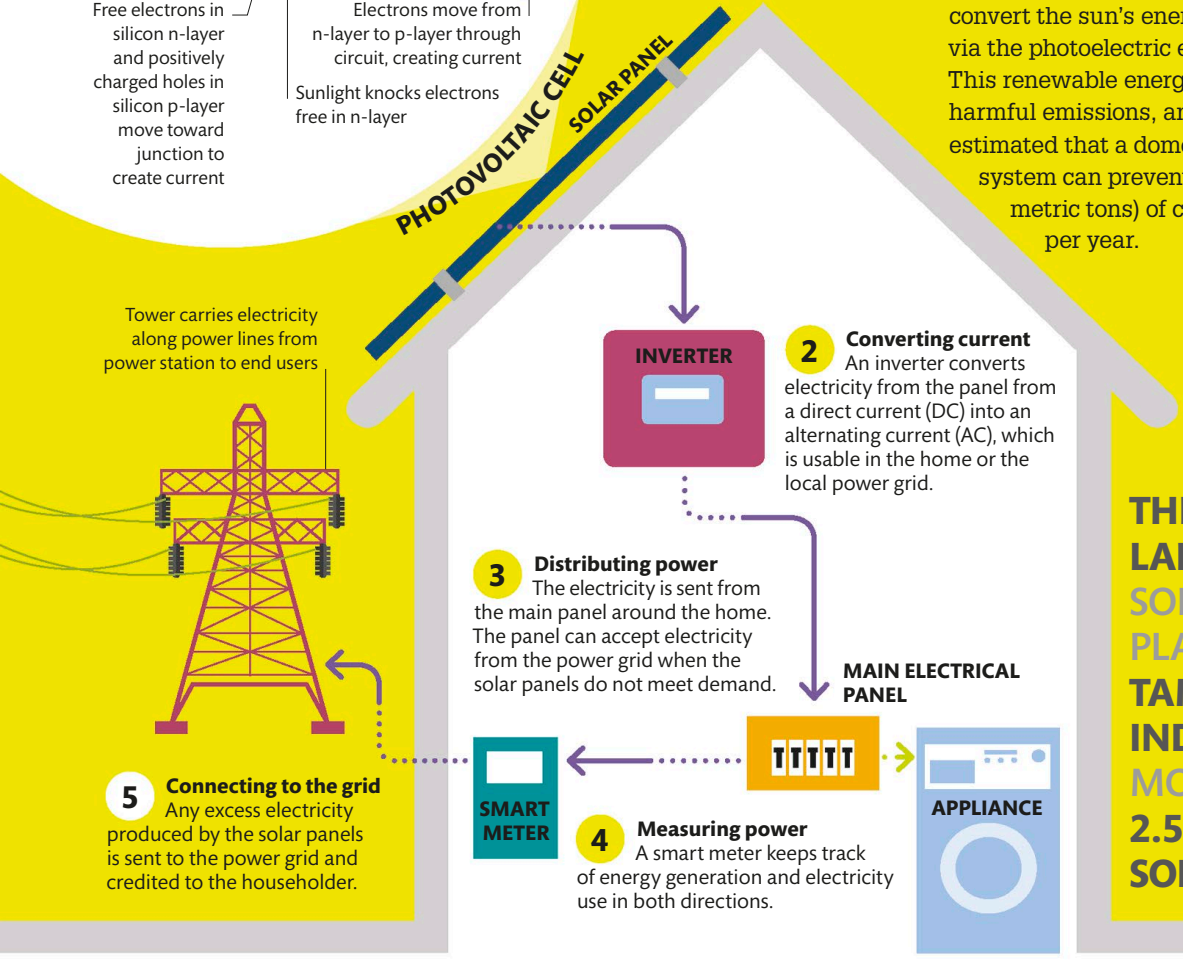


1 Harnessing solar energy
Sunlight comprises tiny energy packages called photons. When photons hit a solar panel, made of dozens of photovoltaic cells, they transfer energy to electrons that then flow via conductors through an external circuit as an electrical current.

Solar power

Energy from sunlight can be harnessed in several ways. Solar collectors use the sun to heat water contained in pipes. Photovoltaic (PV) panels—made of silicon combined, or "doped," with other elements to conduct electricity—convert the sun's energy into electricity via the photoelectric effect (see left). This renewable energy produces no harmful emissions, and it has been estimated that a domestic photovoltaic system can prevent up to 1.9 tons (1.7 metric tons) of carbon emissions per year.



2 Converting current
An inverter converts electricity from the panel from a direct current (DC) into an alternating current (AC), which is usable in the home or the local power grid.

3 Distributing power
The electricity is sent from the main panel around the home. The panel can accept electricity from the power grid when the solar panels do not meet demand.

4 Measuring power
A smart meter keeps track of energy generation and electricity use in both directions.

5 Connecting to the grid
Any excess electricity produced by the solar panels is sent to the power grid and credited to the householder.

THE WORLD'S LARGEST SOLAR POWER PLANT IS IN TAMIL NADU, INDIA, WITH MORE THAN 2.5 MILLION SOLAR PANELS

Solar power and bioenergy

The power of the sun can be used, on various scales, to heat water directly or to generate large amounts of electricity using photovoltaic cells. Biomass—organic material derived from plants or animals—can also be a valuable source of energy.